Numerical Analysis – Homework #4

Utah State University Fall 2026

1.

A common problem in surveying is to determine the altitudes of a series of points with respect to some reference point. The measurements are subject to error, and so multiple measurements are made of each altitude: one with respect to the reference point and others with respect to all of the other points. This results in an overdetermined linear system, which is solved in the least-squares sense to smooth out the errors.

Suppose there are four points whose altitudes x_1, x_2, x_3, and x_4 are to be determined, and that the following measurements are taken:

$$\begin{aligned} x_1 &= 2.95, & x_2 &= 2.95, & x_3 &= -1.45, & x_4 &= -1.45, \\ x_1 - x_2 &= 1.23, & x_1 - x_3 &= 4.45, & x_1 - x_4 &= 1.61, & & \\ & & x_2 - x_3 &= 3.21, & x_2 - x_4 &= 0.45, & & \\ & & & & x_3 - x_4 &= -2.75. & & \end{aligned}$$

It is obvious that these ten equations in four unknowns cannot be satisfied exactly.

Set up the matrix A and the vector b for the associated least-squares problem. Form the matrix A^TA and the vector A^Tb for the normal equation. Obtain the solution of the least-squares problem by solving the normal equation using Cholesky decomposition (you may use MATLAB's `chol` for this). Determine $\kappa_2(A^TA)$, the 2-norm condition number of A^TA (MATLAB's `cond` can be used for this). Print out and hand in A, b, A^TA, A^Tb, $\kappa_2(A^TA)$, and your computed solution of the least-squares problem, together with a brief explanation of why we can safely obtain the solution using the normal equation in this case.

Remark: The easiest way to produce the solution of the least-squares problem is just to use MATLAB's "backslash" command. On an overdetermined system $Ax = b$, typing `A\b` produces the least-squares solution by computing the QR decomposition of A using Householder transformations with column pivoting. Thus you can check the solution that you compute using the normal equation and Cholesky decomposition just by typing `A\b`.

2.

(Linear Least Squares: Normal Equations) Consider the overdetermined system

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}, \qquad b = \begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix}.$$

(a) Form the normal equations $A^TAx = A^Tb$.

(b) Solve the normal equations to obtain the least–squares solution $x^* = (c_0, c_1)^T$.

(c) Write down the best-fit line $y = c_0 + c_1 x$.

3.

(Least Squares via QR Factorization) Let

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 4 \end{pmatrix}, \qquad b = \begin{pmatrix} 6 \\ 3 \\ 7 \end{pmatrix}.$$

(a) Compute the reduced QR factorization $A = QR$ using the Gram–Schmidt method.

(b) Solve the least squares problem by solving $Rx = Q^T b$.

(c) Compare your result with the normal-equation solution. Are they the same? Which method is more numerically stable?

4.

(Gradient Descent for Least Squares) Consider minimizing

$$f(x) = \frac{1}{2}\|Ax - b\|_2^2, \qquad A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

(a) Compute $\nabla f(x)$.

(b) Using the initial guess $x_0 = (0, 0)^T$ and step size $\alpha = 0.25$, compute one iteration $x_1 = x_0 - \alpha \nabla f(x_0)$.

(c) Compute x_2 using the same step size.

5.

(Conjugate Gradient Method) Solve the system $Ax = b$ using **two iterations** of the Conjugate Gradient (CG) method, where

$$A = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \qquad x_0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Compute:

(a) The initial residual r_0 and direction p_0.

(b) The step size α_0, the new iterate x_1, and residual r_1.

(c) The coefficient β_1 and the new search direction p_1.

(d) Perform the second iteration to obtain x_2.

6.

(Condition Number and Least-Squares Stability) Let

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4.01 \end{pmatrix}.$$

(a) Compute the condition number $\kappa_2(A)$.

(b) Explain why solving $A^T Ax = A^T b$ is ill-conditioned even if A is only mildly ill-conditioned.

(c) Briefly explain why QR-based least squares is preferred in practice.